# A High-Radix and Parametric Cellular Communication System

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**Title:** A High-Radix and Parametric Cellular Communication System

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## Cross-Reference to Related Applications

This application is related to the inventions described in the U.S. Patent Applications for "A Method and System for High-Radix Physical Layer Data Encoding and Representation" and "A Protocol Stack and System for Parametric Data Networking." The present invention applies these principles to the unique environment of cellular communication, the contents of which are hereby incorporated by reference in their entirety.

## Abstract

A cellular communication system utilizing a high-radix physical layer for encoding and transmitting abstract symbols over radio frequency (RF) channels, and a parametric networking protocol for end-to-end symbolic data transport. The system comprises base stations and user equipment configured to transmit and receive data as sequences of N discrete, distinguishable RF states (where N greater than 2), each state directly mapping to an abstract symbol (e.g., a numerical value or mathematical operator). This replaces traditional binary RF modulation. Furthermore, the system is designed to encapsulate and transport parametric data structures representing unevaluated mathematical expressions, allowing for the

end-to-end transmission of perfect-precision information across the wireless cellular network segment, from core network to mobile device, without loss of fidelity due to intermediate binary conversion or processing.

## Background of the Invention

Modern cellular networks are marvels of engineering, providing high-speed wireless connectivity. However, at their core, they remain binary systems. Radio frequency (RF) signals are modulated to carry bits, and all higher-level data, regardless of its inherent complexity or required precision, must be reduced to a bit-stream for transmission. This process involves approximations (e.g., converting a high-precision number to a fixed-width floating-point value) that result in a loss of fidelity before the data even enters the wireless channel. Moreover, the network infrastructure often processes and re-encapsulates these approximated binary data, perpetuating the loss.

## Summary of the Invention

The present invention introduces a cellular communication system that operates on a high-radix physical layer and a parametric networking protocol, enabling the native, end-to-end transport of symbolic and perfect-precision data over wireless channels. This fundamentally alters how information is represented and handled within the cellular environment.

The system comprises: (1) **High-Radix RF Transceivers:** Base stations and user equipment are equipped with transceivers capable of generating and reliably distinguishing N discrete RF states (N greater than 2). These states could be, for example, distinct frequencies, phase shifts, or amplitude levels that directly map to abstract symbols rather than binary digits. (2) **Parametric Protocol Stack:** The network utilizes a parametric protocol stack, similar to that described in related applications, extending it into the wireless domain. Data is encapsulated into parametric packets where the payload is an unevaluated mathematical expression or symbolic representation. (3) **Symbolic Channel Management:** Signaling for channel access, resource allocation, and handover procedures can also be conducted using high-radix or symbolic commands, reducing overhead and increasing efficiency.

This architecture allows a mobile device to receive data such as a parametric expression for a number directly, with its full, intrinsic precision, bypassing the limitations imposed by binary conversions. This is particularly valuable for mobile applications requiring high

computational accuracy, such as on-device AI, augmented reality, or secure financial transactions.

## Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

## Brief Description of the Drawings

**FIG. 1** is a conceptual diagram illustrating the high-radix and parametric data flow in a cellular communication system.

**FIG. 2** is a block diagram of a base station or user equipment transceiver incorporating a high-radix RF physical layer.

## Detailed Description of the Invention

Referring to **FIG. 1**, a high-radix and parametric cellular communication system (100) is shown. A source (e.g., a server in the core network) generates a symbolic data structure (110). This structure is encapsulated into a parametric packet (120) and transmitted across the core network (which may also be parametric). A base station (130) receives this parametric packet. Instead of converting it to a binary bit-stream for wireless transmission, the base station's RF transceiver (135) modulates the RF carrier to directly represent the high-radix symbols of the packet's payload and header. This high-radix RF signal (140) is transmitted to a user equipment (UE) (150). The UE's RF transceiver (155) detects and decodes the high-radix RF states back into the original parametric packet (120), delivering the perfectly precise symbolic data structure (110) to the mobile application.

**FIG. 2** illustrates a block diagram of a transceiver (200) suitable for a base station or user equipment. It comprises a High-Radix RF Modulator (210) which takes abstract symbols

as input and generates N discrete RF states on the antenna (220). A High-Radix RF Demodulator (230) receives RF signals from the antenna (220), distinguishes the N states, and outputs the corresponding abstract symbols. This transceiver operates directly on symbolic data without intermediate binary conversion.

# Claims

What is claimed is:

1. A cellular communication system, comprising:
    - a base station, comprising:
        - a high-radix RF modulator configured to receive a sequence of abstract symbols and generate a corresponding sequence of N discrete, distinguishable radio frequency (RF) states, where N is an integer greater than two;
        - an antenna communicatively coupled to the high-radix RF modulator for transmitting said sequence of RF states; and
    - user equipment, comprising:
        - an antenna for receiving said sequence of RF states;
        - a high-radix RF demodulator configured to detect and distinguish the N discrete RF states and generate a corresponding sequence of abstract symbols.
2. The cellular communication system of claim 1, wherein the abstract symbols represent numerical values or mathematical operators.
3. The cellular communication system of claim 1, wherein data transmitted between the base station and user equipment is encapsulated in parametric packets, each packet containing a payload that is an unevaluated mathematical expression.
4. The cellular communication system of claim 3, wherein the high-radix RF modulator is configured to directly modulate the parametric packet's payload without intermediate binary conversion.
5. The cellular communication system of claim 1, wherein control signaling for channel access, resource allocation, or handover procedures is performed using said high-radix abstract symbols.

6. A method for cellular communication, comprising:

    - at a base station, receiving a data structure representing an abstract symbol;
    - generating, based on said abstract symbol, one of N discrete, distinguishable radio frequency (RF) states, where N is an integer greater than two;
    - transmitting said RF state over a wireless cellular channel;
    - at user equipment, receiving said RF state;
    - distinguishing said RF state from the other N-1 possible RF states; and
    - generating, based on the distinguished RF state, the original abstract symbol.

7. The method of claim 6, wherein the data structure received at the base station is part of a parametric packet payload that represents an unevaluated mathematical expression.

8. The method of claim 6, further comprising managing cellular network resources using commands encoded as high-radix abstract symbols.

# Drawings

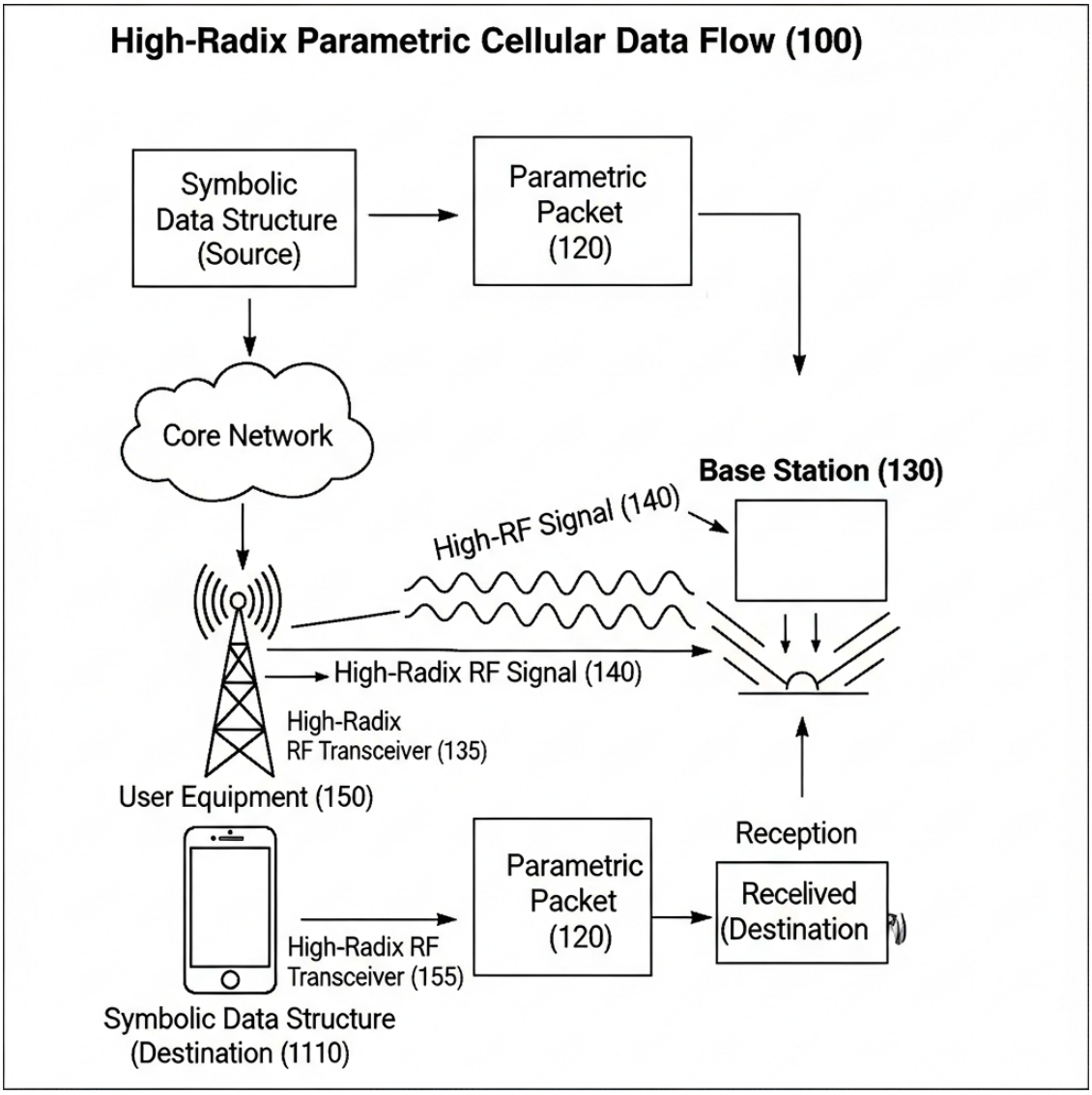



Figure 1: Conceptual diagram illustrating the high-radix and parametric data flow in a cellular communication system.

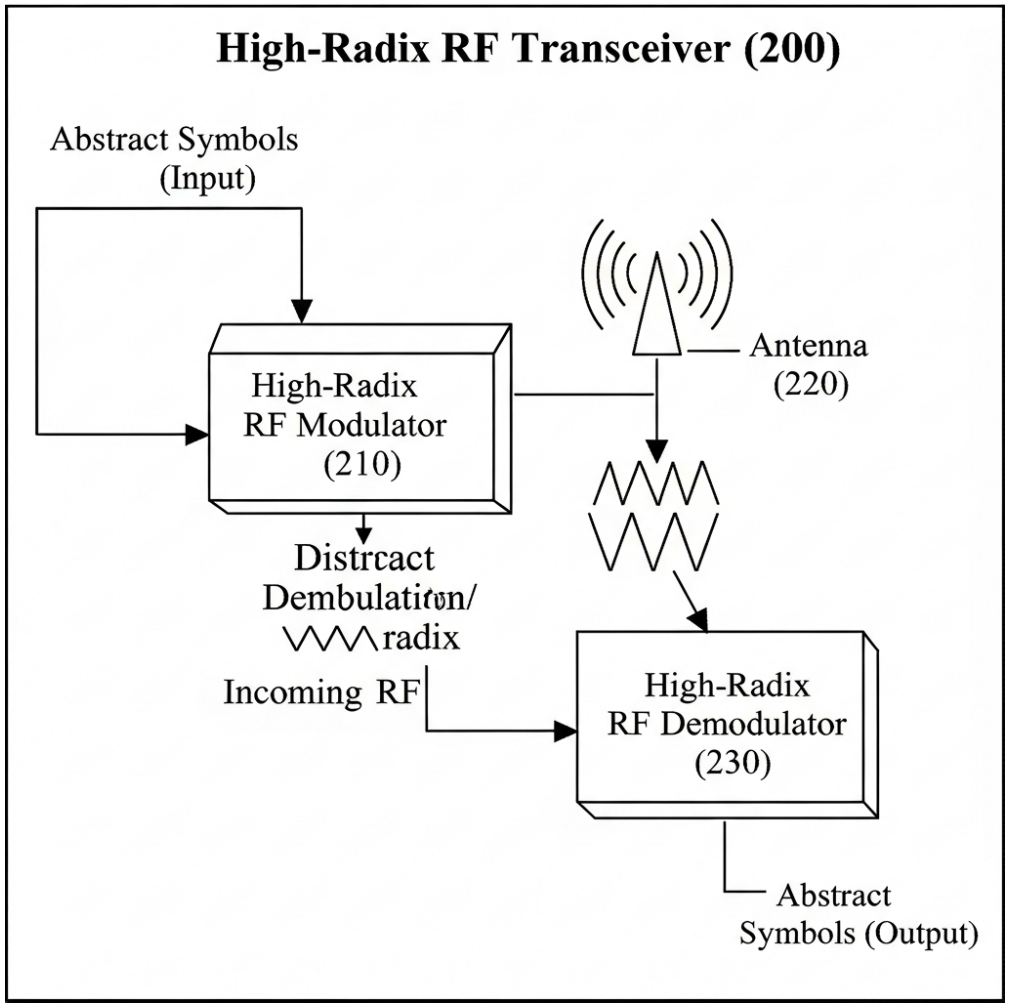



Figure 2: Block diagram of a base station or user equipment transceiver incorporating a high-radix RF physical layer.